Paper - I Set - A

1. The contrapositive of $p \rightarrow q$ is:
   A. $q \rightarrow p$
   B. $\neg p \rightarrow \neg q$
   C. $\neg q \rightarrow \neg p$
   D. $p \rightarrow \neg q$

2. How many subsets does a set with 7 elements have?
   A. 7
   B. 14
   C. 49
   D. 128

3. If $|A|=20$, $|B|=15$ and $|A \cap B|=6$, then $|A \cup B|$ is:
   A. 29
   B. 35
   C. 41
   D. 11

4. If a set has 6 elements, the number of permutations of all its elements is:
   A. 36
   B. 120
   C. 360
   D. 720

5. The number of ways of choosing 3 objects from 8 objects is:
   A. 24
   B. 56
   C. 336
   D. 512

6. For a finite set A with $|A|=4$ and a set B with $|B|=3$, the number of functions $A \rightarrow B$ is:
   A. 12
   B. 24
   C. 64
   D. 81

7. The number of onto functions from a 3-element set to a 2-element set is:
   A. 2
   B. 4
   C. 6
   D. 8

8. If a permutation of 5 distinct symbols is required to have two specified symbols adjacent, the number of such permutations is:
   A. 24
   B. 36
   C. 48
   D. 60

9. How many 5-member committees can be formed from 10 people if two particular people cannot both be selected?
   A. 126
   B. 196
   C. 210
   D. 252

10. The recurrence $T(n)=2T(n/2)+n$, with $T(1)=\Theta(1)$, has asymptotic order:
    A. $\Theta(\log n)$
    B. $\Theta(n)$
    C. $\Theta(n \log n)$
    D. $\Theta(n^2)$

11. If a graph is bipartite, it cannot contain:
    A. a path
    B. an even cycle
    C. an odd cycle
    D. an isolated vertex

12. How many binary strings of length 10 contain exactly four 1s with no two 1s consecutive?
    A. 15
    B. 20
    C. 35
    D. 56

13. For the recurrence $T(n)=T(n-1)+n$ with $T(1)=1$, $T(n)$ is:
    A. $\Theta(\log n)$
    B. $\Theta(n)$
    C. $\Theta(n \log n)$
    D. $\Theta(n^2)$

14. A simple connected planar graph with $V=8$ and $E=12$ has number of faces equal to:
    A. 4
    B. 5
    C. 6
    D. 7

15. A DFA and an NFA recognize:
    A. different classes of regular languages
    B. exactly the same class of regular languages
    C. only finite languages
    D. only context-free languages

16. Mathematical induction is most directly used to prove statements about:
    A. real numbers only
    B. natural numbers
    C. complex numbers only
    D. irrational numbers only

17. A connected graph with n vertices and $n-1$ edges is necessarily a:
    A. cycle
    B. tree
    C. complete graph
    D. multigraph

18. Which traversal of a graph uses a queue in its standard implementation?
    A. DFS
    B. BFS
    C. Backtracking
    D. Topological sort

19. A connected graph has 12 vertices and 15 edges. The number of edges that must be removed to obtain a spanning tree is:
    A. 2
    B. 3
    C. 4
    D. 15

20. Among the Chomsky hierarchy, Type-2 grammars generate:
    A. regular languages
    B. context-free languages
    C. context-sensitive languages
    D. recursively enumerable languages

21. BCD represents each decimal digit using:
    A. 2 bits
    B. 3 bits
    C. 4 bits
    D. 8 bits

22. A stack follows which discipline?
    A. FIFO
    B. LIFO
    C. random access only
    D. priority order

23. Which addressing mode places the operand value directly in the instruction?
    A. immediate
    B. indirect
    C. indexed
    D. register indirect

24. A subroutine return address is commonly stored in:
    A. a link register or stack
    B. the ALU only
    C. the cache tag
    D. the I/O bus

25. If a CPU has a 32-bit address bus and addresses bytes, its addressable memory space is:
    A. 2^{32} bytes
    B. 2^{31} bytes
    C. 32 GB only if word-addressed
    D. 2^{16} bytes

26. The effective address in base-plus-index addressing is typically:
    A. base + index + displacement
    B. PC − instruction length only
    C. opcode + register number
    D. IR – MAR

27. The two's-complement representation of −5 using 8 bits is:
    A. 00000101
    B. 11111010
    C. 11111011
    D. 10000101

28. Which cache mapping allows a memory block to be placed in any cache line?
    A. direct mapping
    B. fully associative mapping
    C. set mapping with one line per set
    D. none

29. Which register normally holds the address of the next instruction to be fetched?
    A. IR
    B. PC
    C. MAR
    D. MDR

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31. In a basic instruction cycle, instruction fetch occurs before:
    A. instruction execution
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    C. disk formatting
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32. If cache access time is 2 ns, main-memory access is 80 ns, and hit ratio is 0.95 (assuming serial lookup), the approximate average access time is:
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    B. 82 ns
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    A. CPU to execute floating-point operations faster
    B. I/O devices to transfer data to/from memory with limited CPU intervention
    C. cache to replace RAM
    D. the ALU to access disk directly

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    A. eliminate all page faults
    B. allow programs to use an address space larger than physical memory
    C. make registers larger
    D. remove the need for secondary storage

35. The primary purpose of an instruction register is to:
    A. hold the current instruction for decoding/control
    B. hold the next memory address
    C. store all arithmetic results
    D. control disk rotation

36. A CPU has a 5-stage pipeline with stage times 2, 3, 1, 4 and 2 ns. Ignoring pipeline register overhead, the minimum clock period is:
    A. 2 ns
    B. 3 ns
    C. 4 ns
    D. 12 ns

37. A direct-mapped cache has 64 lines and a block size of 16 bytes. For a 32-bit byte address, the number of bits used for the block offset is:
    A. 2
    B. 4
    C. 6
    D. 8

38. A pipeline has an ideal CPI of 1. A branch instruction occurs every 5 instructions, and 30% of branches incur a 3-cycle penalty. The effective CPI is:
    A. 1.06
    B. 1.12
    C. 1.18
    D. 1.30

39. If a 4-way set-associative cache has 128 sets and each block is 32 bytes, a 32-bit byte address uses how many tag bits?
    A. 16
    B. 17
    C. 18
    D. 19

40. A 1 MB cache uses 64-byte blocks and is 4-way set associative. The number of cache sets is:
    A. 4096
    B. 8192
    C. 16384
    D. 65536

41. A priority queue removes elements according to:
    A. insertion order necessarily
    B. priority
    C. alphabetical order only
    D. memory address

42. UNION-FIND is particularly useful for:
    A. string sorting
    B. disjoint-set connectivity problems
    C. matrix multiplication
    D. expression parsing

43. AVL trees maintain balance using:
    A. hashing
    B. rotations
    C. BFS only
    D. heapification

44. In a max-heap, every parent key is:
    A. Not greater than its children
    B. Not less than its children
    C. Equal to all descendants
    D. Unrelated to children

45. The time complexity of BFS using adjacency lists is:
    A. $\Theta(V)$ regardless of E
    B. $\Theta(E^2)$
    C. $\Theta(V+E)$
    D. $\Theta(VE^2)$

46. Kruskal's algorithm is a:
    A. divide-and-conquer algorithm
    B. greedy algorithm
    C. dynamic programming algorithm
    D. backtracking algorithm

47. Quicksort has worst-case complexity:
    A. $\Theta(\log n)$
    B. $\Theta(n)$
    C. $\Theta(n \log n)$
    D. $\Theta(n^2)$

48. If a balanced BST contains n nodes, search takes:
A. Θ(1)
B. Θ(log n)
C. Θ(n)
D. Θ(n^2)

49. Deleting a node with two children from a BST is commonly handled by replacing it with its:
A. root only
B. inorder predecessor or successor
C. leftmost leaf only
D. rightmost leaf only

50. Which data structure is most directly suited to implementing function-call recursion?
A. queue
B. stack
C. heap
D. hash table

51. Binary search requires the input array to be:
A. randomized
B. sorted
C. circular
D. hashed

52. The worst-case time complexity of insertion sort is:
A. Θ(log n)
B. Θ(n)
C. Θ(n log n)
D. Θ(n^2)

53. Merge sort has worst-case time complexity:
A. Θ(n)
B. Θ(log n)
C. Θ(n log n)
D. Θ(n^2)

54. A stack operation that removes the top element is:
A. push
B. pop
C. peek only
D. enqueuer

55. For a complete binary tree with n nodes, its height is:
A. Θ(1)
B. Θ(log n)
C. Θ(n)
D. Θ(n log n)

56. Building a binary heap from n arbitrary elements using bottom-up heapify takes:
A. Θ(log n)
B. Θ(n)
C. Θ(n log n)
D. Θ(n^2)

57. Heap sort has worst-case time complexity:
A. Θ(n)
B. Θ(n log n)
C. Θ(n^2)
D. Θ(log n)

58. Which sorting algorithm is stable in its standard comparison-based implementation?
A. heap sort
B. merge sort
C. selection sort
D. quicksort

59. For a sparse graph, an adjacency-list representation generally uses:
A. Θ(V^2) space necessarily
B. Θ(V+E) space
C. Θ(E^2) space
D. Θ(log V) space

60. Union by rank/size primarily keeps:
A. set representatives random
B. disjoint-set trees shallow
C. all sets equal-sized
D. edges sorted

61. The postfix form of (A+B)*(C-D) is:
A. AB+CD-*
B. ABCD+-*
C. AB+*CD-
D. A+B*C-D

62. Deleting the last node of a singly linked list without a tail pointer generally takes:
A. Θ(1)
B. Θ(log n)
C. Θ(n)
D. $\Theta(n^2)$

63. A comparison sorting algorithm that is stable, in-place in its standard form, and has worst-case Θ(n log n) is:
A. heap sort
B. merge sort
C. insertion sort
D. quicksort with arbitrary pivot

64. A graph has V vertices and E edges. If all edge weights are distinct, the MST is:
A. always non-unique
B. unique
C. necessarily a path
D. necessarily a star

65. If a graph contains a negative-weight edge but no negative cycle, which algorithm among the following is appropriate for single-source shortest paths in general?
A. Dijkstra
B. Bellman-Ford
C. BFS only
D. Prim

66. A hash table with open addressing has load factor α. As α approaches 1, expected unsuccessful-search cost generally:
A. decreases to 1
B. increases significantly
C. remains exactly constant
D. becomes zero

67. A binary search tree can have height Θ(n) when keys are inserted in:
A. random order only
B. sorted order
C. strictly alternating order only
D. heap order

68. Which statement about an AVL tree with n nodes is correct?
A. Its height is Θ(n)
B. Its height is Θ(log n)
C. It has exactly n/2 leaves
D. Every node has two children

69. If a binary heap has height h, the number of nodes is:
A. always exactly 2^h
B. between 2^h and 2^(h+1)−1
C. between h and 2h
D. exactly h^2

70. For a connected undirected graph, if DFS encounters a visited vertex that is not the parent of the current vertex, this can indicate:
A. a cycle
B. a minimum spanning tree
C. a topological order
D. a bipartite partition necessarily

71. Which scheduling algorithm is non-preemptive by its basic definition?
A. First Come First Serve
B. Round Robin
C. preemptive priority
D. Shortest Remaining Time First

72. Round Robin scheduling is primarily designed for:
A. batch-only systems
B. time-sharing systems
C. disk formatting
D. deadlock detection

73. The critical-section problem concerns:
A. disk capacity
B. mutual exclusion and coordination when accessing shared data
C. instruction encoding
D. memory addressing only

74. A context switch changes the CPU from:
A. one disk sector to another
B. one file to another
C. one process/thread context to another
D. one page size to another

75. Virtual memory allows:
A. execution of programs whose address space can exceed physical memory
B. elimination of secondary storage
C. all pages to remain in RAM
D. only kernel programs to run

76. A page fault occurs when:
A. a page is in the TLB
B. a process terminates normally
C. a page is not present in physical memory when referenced
D. a file is opened

77. Thrashing refers to:
A. deleting unused pages
B. excessive paging activity with little useful computation
C. CPU overheating
D. deadlock recovery

78. A file system directory primarily maintains:
A. associations between file names and file metadata/references
B. CPU instructions only
C. RAM refresh cycles
D. network routes only

79. A real-time operating system emphasizes:
A. maximum disk fragmentation
B. predictable timing behavior
C. unbounded scheduling delay
D. interactive graphics only

80. The producer-consumer problem is primarily about coordinating:
A. CPU registers
B. file permissions only
C. processes sharing a bounded buffer
D. page tables only

81. In FCFS scheduling, processes P1=24 ms, P2=3 ms, P3=3 ms arrive at time 0 in that order. The average waiting time is:
A. 3 ms
B. 9 ms
C. 17 ms
D. 18 ms

82. Round Robin with time quantum 2 ms is generally considered fairer than FCFS because it:
A. always minimizes average waiting time
B. eliminates context switches
C. is non-preemptive
D. gives each ready process recurring CPU opportunities

83. The four Coffman deadlock conditions are mutual exclusion, hold-and-wait, no preemption, and:
A. circular wait
B. priority inversion
C. starvation
D. paging

84. Banker's algorithm is a deadlock:
A. prevention method
B. avoidance method
C. recovery method
D. detection-only method

85. FIFO page replacement can exhibit Belady's anomaly, meaning:
A. more frames can sometimes produce more page faults
B. fewer frames always produce fewer faults
C. page faults become impossible
D. LRU becomes optimal

86. LRU page replacement is based on the assumption that pages used recently are:
A. likely to be used again soon
B. never used again
C. always dirty
D. stored on disk only

87. External fragmentation is particularly associated with:
A. fixed-size paging only
B. variable-size contiguous memory allocation
C. register allocation
D. CPU pipelining

88. In fixed partitioning, a major source of wasted space inside allocated partitions is mainly:
A. internal fragmentation
B. external fragmentation
C. deadlock
D. page faults

89. The working-set model is used primarily to:
A. control thrashing by tracking recently used pages
B. schedule disk arms
C. detect syntax errors
D. allocate file names

90. A disk scheduling algorithm that services requests in one direction and then reverses is:
A. FCFS
B. SCAN
C. SSTF
D. C-SCAN

91. C-SCAN differs from SCAN because it:
A. always chooses the nearest request
B. is purely random
C. services requests in one direction and returns without servicing on the return path
D. uses CPU bursts

92. A file allocation method using a linked list of blocks has the advantage of:
A. no external fragmentation for file blocks, but poor direct access
B. constant-time direct access
C. no pointer overhead
D. requiring contiguous disk space

93. A system with a single CPU and no I/O can still suffer from:
A. deadlock due to multiple resources such as locks
B. page faults
C. context switches
D. all of the above

94. In a paging system, effective memory access is 100 ns, TLB lookup is 10 ns, TLB hit ratio is 0.9, and a TLB miss requires one extra memory access for the page table. Ignoring page faults, the effective access time is:
A. 100 ns
B. 120 ns
C. 140 ns
D. 180 ns

95. For FIFO with 3 frames, the reference string 1,2,3,1,4,2 produces how many page faults assuming initial empty frames?
A. 4
B. 5
C. 6
D. 3

96. The classic producer-consumer solution using semaphores typically uses:
A. only one mutex and no counting
B. only a page table
C. mutex plus empty and full counting semaphores
D. only busy waiting

97. If a scheduler uses aging, its primary purpose is to:
A. increase starvation
B. reduce starvation of low-priority processes
C. eliminate context switches
D. reduce memory size

98. In a system using segmentation, a logical address typically consists of:
A. segment number and offset
B. page number only
C. frame number only
D. disk cylinder and sector

99. A virtual address translation uses a 4-level page table. If each level indexes 9 bits and the page offset is 12 bits, the virtual address size is:
A. 36 bits
B. 39 bits
C. 48 bits
D. 57 bits

100. A 48-bit virtual address with 4 KB pages has how many virtual pages?
A. 2^12
B. 2^36
C. 2^48
D. 2^24

Paper - II Set - B

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    C. $\Theta(n^2)$
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56. Deleting a node with two children from a BST is commonly handled by replacing it with its:
    A. root only
    B. inorder predecessor or successor
    C. leftmost leaf only
    D. rightmost leaf only

57. Building a binary heap from n arbitrary elements using bottom-up heapify takes:
    A. $\Theta(\log n)$
    B. $\Theta(n)$
    C. $\Theta(n \log n)$
    D. $\Theta(n^2)$

58. A graph has V vertices and E edges. If all edge weights are distinct, the MST is:
    A. always non-unique
    B. unique
    C. necessarily a path
    D. necessarily a star

59. Which sorting algorithm is stable in its standard comparison-based implementation?
    A. heap sort
    B. merge sort
    C. selection sort
    D. quicksort
60. The postfix form of (A+B)*(C-D) is:
    A. AB+CD-*
    B. ABCD+-*
    C. AB+*CD-
    D. A+B*C-D

61. Deleting the last node of a singly linked list without a tail pointer generally takes:
    A. Θ(1)
    B. Θ(log n)
    C. Θ(n)
    D. Θ(n²)

62. A comparison sorting algorithm that is stable, in-place in its standard form, and has worst-case Θ(n log n) is:
    A. heap sort
    B. merge sort
    C. insertion sort
    D. quicksort with arbitrary pivot

63. A hash table with open addressing has load factor α. As α approaches 1, expected unsuccessful-search cost generally:
    A. decreases to 1
    B. increases significantly
    C. remains exactly constant
    D. becomes zero

64. For a sparse graph, an adjacency-list representation generally uses:
    A. Θ(V²) space necessarily
    B. Θ(V+E) space
    C. Θ(E²) space
    D. Θ(log V) space

65. Union by rank/size primarily keeps:
    A. set representatives random
    B. disjoint-set trees shallow
    C. all sets equal-sized
    D. edges sorted
66. If a binary heap has height h, the number of nodes is:
    A. always exactly 2^h
    B. between 2^h and 2^(h+1)−1
    C. between h and 2h
    D. exactly h²

67. If a graph contains a negative-weight edge but no negative cycle, which algorithm among the following is appropriate for single-source shortest paths in general?
    A. Dijkstra
    B. Bellman-Ford
    C. BFS only
    D. Prim

68. A binary search tree can have height Θ(n) when keys are inserted in:
    A. random order only
    B. sorted order
    C. strictly alternating order only
    D. heap order

69. Which statement about an AVL tree with n nodes is correct?
    A. Its height is Θ(n)
    B. Its height is Θ(log n)
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B. a minimum spanning tree
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71. Thrashing refers to:
A. deleting unused pages
B. excessive paging activity with little useful computation
C. CPU overheating
D. deadlock recovery

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A. First Come First Serve
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B. one file to another
C. one process/thread context to another
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80. Round Robin with time quantum 2 ms is generally considered fairer than FCFS because it:
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B. eliminates context switches
C. is non-preemptive
D. gives each ready process recurring CPU opportunities

81. The four Coffman deadlock conditions are mutual exclusion, hold-and-wait, no preemption, and:
A. circular wait
B. priority inversion
C. starvation
D. paging

82. FIFO page replacement can exhibit Belady's anomaly, meaning:
A. more frames can sometimes produce more page faults
B. fewer frames always produce fewer faults
C. page faults become impossible
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86. LRU page replacement is based on the assumption that pages used recently are:
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B. never used again
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87. External fragmentation is particularly associated with:
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B. variable-size contiguous memory allocation
C. register allocation
D. CPU pipelining

88. A 48-bit virtual address with 4 KB pages has how many virtual pages?
A. 2^12
B. 2^36
C. 2^48
D. 2^24

89. A disk scheduling algorithm that services requests in one direction and then reverses is:
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B. SCAN
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A. always chooses the nearest request
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B. 5
C. 6
D. 3

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A. increase starvation
B. reduce starvation of low-priority processes
C. eliminate context switches
D. reduce memory size

99. In a system using segmentation, a logical address typically consists of:
A. segment number and offset
B. page number only
C. frame number only
D. disk cylinder and sector

100. A virtual address translation uses a 4-level page table. If each level indexes 9 bits and the page offset is 12 bits, the virtual address size is:
A. 36 bits
B. 39 bits
C. 48 bits
D. 57 bits

Paper - I Set - C

1. If a set has 6 elements, the number of permutations of all its elements is:
   A. 36
   B. 120
   C. 360
   D. 720

2. How many subsets does a set with 7 elements have?
   A. 7
   B. 14
   C. 49
   D. 128

3. If $|A|=20$, $|B|=15$ and $|A \cap B|=6$, then $|A \cup B|$ is:
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   B. 35
   C. 41
   D. 11

4. The number of ways of choosing 3 objects from 8 objects is:
   A. 24
   B. 56
   C. 336
   D. 512

5. Mathematical induction is most directly used to prove statements about:
   A. real numbers only
   B. natural numbers
   C. complex numbers only
   D. irrational numbers only

6. How many 5-member committees can be formed from 10 people if two particular people cannot both be selected?
   A. 126
   B. 196
   C. 210
   D. 252

7. The contrapositive of $p \rightarrow q$ is:
   A. $q \rightarrow p$
   B. $\neg p \rightarrow \neg q$
   C. $\neg q \rightarrow \neg p$
   D. $p \rightarrow \neg q$

8. A connected graph with n vertices and $n-1$ edges is necessarily a:
   A. cycle
   B. tree
   C. complete graph
   D. multigraph

9. Which traversal of a graph uses a queue in its standard implementation?
   A. DFS
   B. BFS
   C. Backtracking
   D. Topological sort

10. For a finite set A with $|A|=4$ and a set B with $|B|=3$, the number of functions $A \rightarrow B$ is:
    A. 12
    B. 24
    C. 64
    D. 81

11. The number of onto functions from a 3-element set to a 2-element set is:
    A. 2
    B. 4
    C. 6
    D. 8

12. If a permutation of 5 distinct symbols is required to have two specified symbols adjacent, the number of such permutations is:
    A. 24
    B. 36
    C. 48
    D. 60
13. Among the Chomsky hierarchy, Type-2 grammars generate:
    A. regular languages
    B. context-free languages
    C. context-sensitive languages
    D. recursively enumerable languages

14. How many binary strings of length 10 contain exactly four 1s with no two 1s consecutive?
    A. 15
    B. 20
    C. 35
    D. 56

15. For the recurrence $T(n)=T(n-1)+n$ with $T(1)=1$, $T(n)$ is:
    A. $\Theta(\log n)$
    B. $\Theta(n)$
    C. $\Theta(n \log n)$
    D. $\Theta(n^2)$

16. A connected graph has 12 vertices and 15 edges. The number of edges that must be removed to obtain a spanning tree is:
    A. 2
    B. 3
    C. 4
    D. 15

17. A simple connected planar graph with $V=8$ and $E=12$ has number of faces equal to:
    A. 4
    B. 5
    C. 6
    D. 7

18. The recurrence $T(n)=2T(n/2)+n$, with $T(1)=\Theta(1)$, has asymptotic order:
    A. $\Theta(\log n)$
    B. $\Theta(n)$
    C. $\Theta(n \log n)$
    D. $\Theta(n^2)$
19. If a graph is bipartite, it cannot contain:
    A. a path
    B. an even cycle
    C. an odd cycle
    D. an isolated vertex

20. A DFA and an NFA recognize:
    A. different classes of regular languages
    B. exactly the same class of regular languages
    C. only finite languages
    D. only context-free languages

21. In a basic instruction cycle, instruction fetch occurs before:
    A. instruction execution
    B. power-on
    C. disk formatting
    D. program compilation

22. A stack follows which discipline?
    A. FIFO
    B. LIFO
    C. random access only
    D. priority order

23. Which addressing mode places the operand value directly in the instruction?
    A. immediate
    B. indirect
    C. indexed
    D. register indirect

24. BCD represents each decimal digit using:
    A. 2 bits
    B. 3 bits
    C. 4 bits
    D. 8 bits

25. The register that holds the currently fetched instruction is the:
    A. IR
    B. PC
    C. ACC
    D. MAR

26. A direct-mapped cache has 64 lines and a block size of 16 bytes. For a 32-bit byte address, the number of bits used for the block offset is:
    A. 2
    B. 4
    C. 6
    D. 8

27. A subroutine return address is commonly stored in:
    A. a link register or stack
    B. the ALU only
    C. the cache tag
    D. the I/O bus

28. If a CPU has a 32-bit address bus and addresses bytes, its addressable memory space is:
    A. 2^{32} bytes
    B. 2^{31} bytes
    C. 32 GB only if word-addressed
    D. 2^{16} bytes

29. The effective address in base-plus-index addressing is typically:
    A. base + index + displacement
    B. PC − instruction length only
    C. opcode + register number
    D. IR – MAR

30. If cache access time is 2 ns, main-memory access is 80 ns, and hit ratio is 0.95 (assuming serial lookup), the approximate average access time is:
    A. 6 ns
    B. 82 ns
    C. 80 ns
    D. 4 ns

31. DMA primarily allows:
    A. CPU to execute floating-point operations faster
    B. I/O devices to transfer data to/from memory with limited CPU intervention
    C. cache to replace RAM
    D. the ALU to access disk directly

32. A pipeline has an ideal CPI of 1. A branch instruction occurs every 5 instructions, and 30% of branches incur a 3-cycle penalty. The effective CPI is:
    A. 1.06
    B. 1.12
    C. 1.18
    D. 1.30

33. A 1 MB cache uses 64-byte blocks and is 4-way set associative. The number of cache sets is:
    A. 4096
    B. 8192
    C. 16384
    D. 65536

34. The main purpose of virtual memory is to:
A. eliminate all page faults
B. allow programs to use an address space larger than physical memory
C. make registers larger
D. remove the need for secondary storage

35. The primary purpose of an instruction register is to:
A. hold the current instruction for decoding/control
B. hold the next memory address
C. store all arithmetic results
D. control disk rotation

36. Which cache mapping allows a memory block to be placed in any cache line?
A. direct mapping
B. fully associative mapping
C. set mapping with one line per set
D. none

37. The two's-complement representation of −5 using 8 bits is:
A. 00000101
B. 11111010
C. 11111011
D. 10000101

38. Which register normally holds the address of the next instruction to be fetched?
A. IR
B. PC
C. MAR
D. MDR

39. A CPU has a 5-stage pipeline with stage times 2, 3, 1, 4 and 2 ns. Ignoring pipeline register overhead, the minimum clock period is:
A. 2 ns
B. 3 ns
C. 4 ns
D. 12 ns

40. If a 4-way set-associative cache has 128 sets and each block is 32 bytes, a 32-bit byte address uses how many tag bits?
A. 16
B. 17
C. 18
D. 19

41. Binary search requires the input array to be:
A. randomized
B. sorted
C. circular
D. hashed

42. The worst-case time complexity of insertion sort is:
A. $\Theta(\log n)$
B. $\Theta(n)$
C. $\Theta(n \log n)$
D. $\Theta(n^2)$

43. Merge sort has worst-case time complexity:
A. $\Theta(n)$
B. $\Theta(\log n)$
C. $\Theta(n \log n)$
D. $\Theta(n^2)$

44. A stack operation that removes the top element is:
A. push
B. pop
C. peek only
D. enqueuer

45. A priority queue removes elements according to:
    A. insertion order necessarily
    B. priority
    C. alphabetical order only
    D. memory address

46. UNION-FIND is particularly useful for:
    A. string sorting
    B. disjoint-set connectivity problems
    C. matrix multiplication
    D. expression parsing

47. AVL trees maintain balance using:
    A. hashing
    B. rotations
    C. BFS only
    D. heapification

48. In a max-heap, every parent key is:
    A. Not greater than its children
    B. Not less than its children
    C. Equal to all descendants
    D. Unrelated to children

49. The time complexity of BFS using adjacency lists is:
    A. $\Theta(V)$ regardless of E
    B. $\Theta(E^2)$
    C. $\Theta(V+E)$
    D. $\Theta(VE^2)$

50. Kruskal's algorithm is a:
    A. divide-and-conquer algorithm
    B. greedy algorithm
    C. dynamic programming algorithm
    D. backtracking algorithm

51. Quicksort has worst-case complexity:
    A. $\Theta(\log n)$
    B. $\Theta(n)$
    C. $\Theta(n \log n)$
    D. $\Theta(n^2)$

52. For a complete binary tree with n nodes, its height is:
    A. $\Theta(1)$
    B. $\Theta(\log n)$
    C. $\Theta(n)$
    D. $\Theta(n \log n)$

53. Which data structure is most directly suited to implementing function-call recursion?
    A. queue
    B. stack
    C. heap
    D. hash table

54. If a balanced BST contains n nodes, search takes:
    A. $\Theta(1)$
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Paper - I Set - D

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   D. 8

10. Which traversal of a graph uses a queue in its standard implementation?
    A. DFS
    B. BFS
    C. Backtracking
    D. Topological sort

11. How many 5-member committees can be formed from 10 people if two particular people cannot both be selected?
    A. 126
    B. 196
    C. 210
    D. 252

12. If a graph is bipartite, it cannot contain:
    A. a path
    B. an even cycle
    C. an odd cycle
    D. an isolated vertex

13. Among the Chomsky hierarchy, Type-2 grammars generate:
    A. regular languages
    B. context-free languages
    C. context-sensitive languages
    D. recursively enumerable languages

14. If a permutation of 5 distinct symbols is required to have two specified symbols adjacent, the number of such permutations is:
    A. 24
    B. 36
    C. 48
    D. 60

15. How many binary strings of length 10 contain exactly four 1s with no two 1s consecutive?
    A. 15
    B. 20
    C. 35
    D. 56

16. The contrapositive of $p \rightarrow q$ is:
    A. $q \rightarrow p$
    B. $\neg p \rightarrow \neg q$
    C. $\neg q \rightarrow \neg p$
    D. $p \rightarrow \neg q$

17. For the recurrence $T(n)=T(n-1)+n$ with $T(1)=1$, $T(n)$ is:
    A. $\Theta(\log n)$
    B. $\Theta(n)$
    C. $\Theta(n \log n)$
    D. $\Theta(n^2)$

18. A simple connected planar graph with $V=8$ and $E=12$ has number of faces equal to:
    A. 4
    B. 5
    C. 6
    D. 7

19. A DFA and an NFA recognize:
    A. different classes of regular languages
    B. exactly the same class of regular languages
    C. only finite languages
    D. only context-free languages

20. A connected graph has 12 vertices and 15 edges. The number of edges that must be removed to obtain a spanning tree is:
    A. 2
    B. 3
    C. 4
    D. 15

21. The register that holds the currently fetched instruction is the:
    A. IR
    B. PC
    C. ACC
    D. MAR

22. Which addressing mode places the operand value directly in the instruction?
    A. immediate
    B. indirect
    C. indexed
    D. register indirect

23. A subroutine return address is commonly stored in:
    A. a link register or stack
    B. the ALU only
    C. the cache tag
    D. the I/O bus

24. If a CPU has a 32-bit address bus and addresses bytes, its addressable memory space is:
    A. 2^{32} bytes
    B. 2^{31} bytes
    C. 32 GB only if word-addressed
    D. 2^{16} bytes
25. In a basic instruction cycle, instruction fetch occurs before:
    A. instruction execution
    B. power-on
    C. disk formatting
    D. program compilation

26. BCD represents each decimal digit using:
    A. 2 bits
    B. 3 bits
    C. 4 bits
    D. 8 bits

27. A stack follows which discipline?
    A. FIFO
    B. LIFO
    C. random access only
    D. priority order

28. The effective address in base-plus-index addressing is typically:
    A. base + index + displacement
    B. PC − instruction length only
    C. opcode + register number
    D. IR – MAR

29. The two's-complement representation of −5 using 8 bits is:
    A. 00000101
    B. 11111010
    C. 11111011
    D. 10000101

30. Which cache mapping allows a memory block to be placed in any cache line?
    A. direct mapping
    B. fully associative mapping
    C. set mapping with one line per set
    D. none

31. A 1 MB cache uses 64-byte blocks and is 4-way set associative. The number of cache sets is:
    A. 4096
    B. 8192
    C. 16384
    D. 65536

32. If cache access time is 2 ns, main-memory access is 80 ns, and hit ratio is 0.95 (assuming serial lookup), the approximate average access time is:
    A. 6 ns
    B. 82 ns
    C. 80 ns
    D. 4 ns

33. DMA primarily allows:
    A. CPU to execute floating-point operations faster
    B. I/O devices to transfer data to/from memory with limited CPU intervention
    C. cache to replace RAM
    D. the ALU to access disk directly

34. Which register normally holds the address of the next instruction to be fetched?
    A. IR
    B. PC
    C. MAR
    D. MDR

35. A direct-mapped cache has 64 lines and a block size of 16 bytes. For a 32-bit byte address, the number of bits used for the block offset is:
A. 2
B. 4
C. 6
D. 8

36. A pipeline has an ideal CPI of 1. A branch instruction occurs every 5 instructions, and 30% of branches incur a 3-cycle penalty. The effective CPI is:
A. 1.06
B. 1.12
C. 1.18
D. 1.30

37. If a 4-way set-associative cache has 128 sets and each block is 32 bytes, a 32-bit byte address uses how many tag bits?
A. 16
B. 17
C. 18
D. 19

38. The main purpose of virtual memory is to:
A. eliminate all page faults
B. allow programs to use an address space larger than physical memory
C. make registers larger
D. remove the need for secondary storage

39. The primary purpose of an instruction register is to:
A. hold the current instruction for decoding/control
B. hold the next memory address
C. store all arithmetic results
D. control disk rotation

40. A CPU has a 5-stage pipeline with stage times 2, 3, 1, 4 and 2 ns. Ignoring pipeline register overhead, the minimum clock period is:
A. 2 ns
B. 3 ns
C. 4 ns
D. 12 ns

41. A stack operation that removes the top element is:
A. push
B. pop
C. peek only
D. enqueuer

42. Binary search requires the input array to be:
A. randomized
B. sorted
C. circular
D. hashed

43. The worst-case time complexity of insertion sort is:
A. $\Theta(\log n)$
B. $\Theta(n)$
C. $\Theta(n \log n)$
D. $\Theta(n^2)$

44. Merge sort has worst-case time complexity:
A. $\Theta(n)$
B. $\Theta(\log n)$
C. $\Theta(n \log n)$
D. $\Theta(n^2)$

45. UNION-FIND is particularly useful for:
A. string sorting
B. disjoint-set connectivity problems
C. matrix multiplication
D. expression parsing

46. Which data structure is most directly suited to implementing function-call recursion?
A. queue
B. stack
C. heap
D. hash table

47. A priority queue removes elements according to:
A. insertion order necessarily
B. priority
C. alphabetical order only
D. memory address

48. AVL trees maintain balance using:
A. hashing
B. rotations
C. BFS only
D. heapification

49. The time complexity of BFS using adjacency lists is:
A. $\Theta(V)$ regardless of E
B. $\Theta(E^2)$
C. $\Theta(V+E)$
D. $\Theta(VE^2)$

50. Kruskal's algorithm is a:
A. divide-and-conquer algorithm
B. greedy algorithm
C. dynamic programming algorithm
D. backtracking algorithm

51. In a max-heap, every parent key is:
A. Not greater than its children
B. Not less than its children
C. Equal to all descendants
D. Unrelated to children

52. Quicksort has worst-case complexity:
A. $\Theta(\log n)$
B. $\Theta(n)$
C. $\Theta(n \log n)$
D. $\Theta(n^2)$

53. If a balanced BST contains n nodes, search takes:
A. $\Theta(1)$
B. $\Theta(\log n)$
C. $\Theta(n)$
D. $\Theta(n^2)$

54. Building a binary heap from n arbitrary elements using bottom-up heapify takes:
A. $\Theta(\log n)$
B. $\Theta(n)$
C. $\Theta(n \log n)$
D. $\Theta(n^2)$

55. A comparison sorting algorithm that is stable, in-place in its standard form, and has worst-case $\Theta(n \log n)$ is:
A. heap sort
B. merge sort
C. insertion sort
D. quicksort with arbitrary pivot

56. A graph has V vertices and E edges. If all edge weights are distinct, the MST is:
A. always non-unique
B. unique
C. necessarily a path
D. necessarily a star

57. Heap sort has worst-case time complexity:
A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta(n^2)$
D. $\Theta(\log n)$

58. Which sorting algorithm is stable in its standard comparison-based implementation?
A. heap sort
B. merge sort
C. selection sort
D. quicksort

59. For a sparse graph, an adjacency-list representation generally uses:
A. $\Theta(V^2)$ space necessarily
B. $\Theta(V+E)$ space
C. $\Theta(E^2)$ space
D. $\Theta(\log V)$ space

60. Union by rank/size primarily keeps:
A. set representatives random
B. disjoint-set trees shallow
C. all sets equal-sized
D. edges sorted

61. If a binary heap has height h, the number of nodes is:
A. always exactly 2^h
B. between 2^h and 2^(h+1)−1
C. between h and 2h
D. exactly h^2

62. For a connected undirected graph, if DFS encounters a visited vertex that is not the parent of the current vertex, this can indicate:
A. a cycle
B. a minimum spanning tree
C. a topological order
D. a bipartite partition necessarily

63. Deleting a node with two children from a BST is commonly handled by replacing it with its:
A. root only
B. inorder predecessor or successor
C. leftmost leaf only
D. rightmost leaf only

64. For a complete binary tree with n nodes, its height is:
A. $\Theta(1)$
B. $\Theta(\log n)$
C. $\Theta(n)$
D. $\Theta(n \log n)$

65. The postfix form of (A+B)*(C-D) is:
A. AB+CD-*
B. ABCD+-*
C. AB+*CD-
D. A+B*C-D

66. Deleting the last node of a singly linked list without a tail pointer generally takes:
A. $\Theta(1)$
B. $\Theta(\log n)$
C. $\Theta(n)$
D. $\Theta(n^2)$

67. If a graph contains a negative-weight edge but no negative cycle, which algorithm among the following is appropriate for single-source shortest paths in general?
A. Dijkstra
B. Bellman-Ford
C. BFS only
D. Prim

68. A hash table with open addressing has load factor α. As α approaches 1, expected unsuccessful-search cost generally:
A. decreases to 1
B. increases significantly
C. remains exactly constant
D. becomes zero

69. A binary search tree can have height Θ(n) when keys are inserted in:
A. random order only
B. sorted order
C. strictly alternating order only
D. heap order

70. Which statement about an AVL tree with n nodes is correct?
A. Its height is Θ(n)
B. Its height is Θ(log n)
C. It has exactly n/2 leaves
D. Every node has two children

71. Virtual memory allows:
A. execution of programs whose address space can exceed physical memory
B. elimination of secondary storage
C. all pages to remain in RAM
D. only kernel programs to run

72. Which scheduling algorithm is non-preemptive by its basic definition?
A. First Come First Serve
B. Round Robin
C. preemptive priority
D. Shortest Remaining Time First

73. Round Robin scheduling is primarily designed for:
A. batch-only systems
B. time-sharing systems
C. disk formatting
D. deadlock detection

74. The critical-section problem concerns:
A. disk capacity
B. mutual exclusion and coordination when accessing shared data
C. instruction encoding
D. memory addressing only

75. Thrashing refers to:
A. deleting unused pages
B. excessive paging activity with little useful computation
C. CPU overheating
D. deadlock recovery

76. A file system directory primarily maintains:
A. associations between file names and file metadata/references
B. CPU instructions only
C. RAM refresh cycles
D. network routes only

77. A real-time operating system emphasizes:
A. maximum disk fragmentation
B. predictable timing behavior
C. unbounded scheduling delay
D. interactive graphics only

78. The producer-consumer problem is primarily about coordinating:
A. CPU registers
B. file permissions only
C. processes sharing a bounded buffer
D. page tables only

79. In FCFS scheduling, processes P1=24 ms, P2=3 ms, P3=3 ms arrive at time 0 in that order. The average waiting time is:
A. 3 ms
B. 9 ms
C. 17 ms
D. 18 ms

80. Round Robin with time quantum 2 ms is generally considered fairer than FCFS because it:
A. always minimizes average waiting time
B. eliminates context switches
C. is non-preemptive

D. gives each ready process recurring CPU opportunities

81. The four Coffman deadlock conditions are mutual exclusion, hold-and-wait, no preemption, and:
A. circular wait
B. priority inversion
C. starvation
D. paging

82. Banker's algorithm is a deadlock:
A. prevention method
B. avoidance method
C. recovery method
D. detection-only method

83. FIFO page replacement can exhibit Belady's anomaly, meaning:
A. more frames can sometimes produce more page faults
B. fewer frames always produce fewer faults
C. page faults become impossible
D. LRU becomes optimal

84. LRU page replacement is based on the assumption that pages used recently are:
A. likely to be used again soon
B. never used again
C. always dirty
D. stored on disk only

85. External fragmentation is particularly associated with:
A. fixed-size paging only
B. variable-size contiguous memory allocation
C. register allocation
D. CPU pipelining

86. A context switch changes the CPU from:
A. one disk sector to another
B. one file to another
C. one process/thread context to another
D. one page size to another

87. A page fault occurs when:
A. a page is in the TLB
B. a process terminates normally
C. a page is not present in physical memory when referenced
D. a file is opened

88. In a system using segmentation, a logical address typically consists of:
A. segment number and offset
B. page number only
C. frame number only
D. disk cylinder and sector

89. A virtual address translation uses a 4-level page table. If each level indexes 9 bits and the page offset is 12 bits, the virtual address size is:
A. 36 bits
B. 39 bits
C. 48 bits
D. 57 bits

90. In fixed partitioning, a major source of wasted space inside allocated partitions is mainly:
A. internal fragmentation
B. external fragmentation
C. deadlock
D. page faults

91. The working-set model is used primarily to:
A. control thrashing by tracking recently used pages
B. schedule disk arms
C. detect syntax errors
D. allocate file names

92. A disk scheduling algorithm that services requests in one direction and then reverses is:
A. FCFS
B. SCAN
C. SSTF
D. C-SCAN

93. C-SCAN differs from SCAN because it:
A. always chooses the nearest request
B. is purely random
C. services requests in one direction and returns without servicing on the return path
D. uses CPU bursts

94. For FIFO with 3 frames, the reference string 1,2,3,1,4,2 produces how many page faults assuming initial empty frames?
A. 4
B. 5
C. 6
D. 3

95. A file allocation method using a linked list of blocks has the advantage of:
A. no external fragmentation for file blocks, but poor direct access
B. constant-time direct access
C. no pointer overhead
D. requiring contiguous disk space

96. A 48-bit virtual address with 4 KB pages has how many virtual pages?
A. 2^12
B. 2^36
C. 2^48
D. 2^24

97. A system with a single CPU and no I/O can still suffer from:
A. deadlock due to multiple resources such as locks
B. page faults
C. context switches
D. all of the above

98. In a paging system, effective memory access is 100 ns, TLB lookup is 10 ns, TLB hit ratio is 0.9, and a TLB miss requires one extra memory access for the page table. Ignoring page faults, the effective access time is:
A. 100 ns
B. 120 ns
C. 140 ns
D. 180 ns

99. The classic producer-consumer solution using semaphores typically uses:
A. only one mutex and no counting
B. only a page table
C. mutex plus empty and full counting semaphores
D. only busy waiting

100. If a scheduler uses aging, its primary purpose is to:
A. increase starvation
B. reduce starvation of low-priority processes
C. eliminate context switches
D. reduce memory size